Report on "Matrix Theory Reloaded: A BPS Road to Holography"

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Summary. The paper reinterprets the decoupling limits behind matrix theories on Dp-branes ("MpT" limits) as BPS limits that make the target space non-Lorentzian. It shows that applying such a limit to a curved D-brane background induces the near-horizon limit, and that combining limits gives further holographic pairs, some with non-Lorentzian bulks. After uplifting to M-theory, it conjectures that holography is a DLCQ^n/DLCQ^m correspondence ($m > n$). In this language $\mathrm{AdS}_5/\mathrm{CFT}_4$ is DLCQ^0/DLCQ^1. The paper also shows that undoing a limit is a p-brane generalisation of $T\bar{T}$, and it derives $T\bar{T}$-like flow equations for the DBI action.

1. Principal achievement

The main achievement is a single organising principle that ties together several areas usually studied separately: matrix theory, the near-horizon limit of AdS/CFT, non-Lorentzian (Newton–Cartan-type) geometry, and $T\bar{T}$ deformations. The key technical point is that the BPS decoupling limit on the brane side is the same operation as the near-horizon limit in the bulk. This gives a systematic way to generate holographic pairs: apply one limit for D3/M2/M5, two limits for the D1–D5 system, and get new non-Lorentzian bulk duals as by-products. It also gives a concrete reading of the extra DLCQ that relates matrix theory to AdS/CFT. A further contribution is the set of p-brane $T\bar{T}$ flow equations, read off from the ω-dependence of the DBI action. The simple three-dimensional flow shared by the D2 and M2 branes stands out; it is a natural higher-dimensional analogue of $T\bar{T}$ that may be useful outside string theory. The M0T superalgebra, obtained as an Inönü–Wigner contraction, places the framework on algebraic ground.

2. Principal drawback

The central claim, the DLCQ^n/DLCQ^m conjecture, rests on matching limits and geometric scalings rather than on dynamics. The paper reproduces known dualities and suggests new ones, but it never computes a correlator, spectrum, entropy or protected quantity that would test any of the *new* non-Lorentzian dualities. Several foundations are also left open, as the authors themselves note:

- The non-Lorentzian bulk geometries are not shown to solve the equations of motion or the torsional constraints of an MpT supergravity, which has not itself been formulated in general.
- The ad hoc rescaling of ℓ (and hence of N) used to build these geometries has no clear physical interpretation.
- For $p > 3$, the asymptotic region is strongly coupled or dominated by quantum corrections. This weakens the geometric picture exactly where the extra DLCQ is supposed to live.

- The analysis uses only the Poincaré patch.
- The $T\bar{T}$ flows are derived only for bosonic, abelian actions. Showing that the actual matrix theory (non-abelian SYM) flows to the bulk geometry is therefore still a heuristic argument.

Taken together, the conjecture is an attractive reorganisation of known results, but it is not yet falsifiable.

3. Most worthwhile follow-up

Recommendation: build MpT supergravity for one well-controlled case, take M2T (or the non-relativistic M2 limit of M-theory) as the first case, and use it to check the paper's non-Lorentzian bulk geometries. Then compute their symmetry superalgebra and compare it with the contracted MpT superalgebra of the dual matrix theory.

- **Feasibility.** The ingredients exist. The non-relativistic M2 limit of 11d supergravity and its M2 solutions have already been studied (Lambert et al., Bergshoeff et al.). The paper already contains the method for Inönü–Wigner contractions and gives the candidate geometries explicitly. The work is concrete, largely classical, and suitable for a PhD student or postdoc over 6–12 months.
- **Impact.** This would be the first non-trivial consistency test of the new non-Lorentzian holographic pairs. It would test whether the geometries are true solutions, whether the N-rescaling corresponds to physical charges, and whether bulk isometries match boundary symmetries. Each of these checks either supports or rules out the DLCQ^n/DLCQ^m conjecture. It would also provide the bulk side needed for later dynamical tests, such as two-point functions and thermodynamics at finite temperature.
- **Comparison with alternatives.** Extending the p-brane $T\bar{T}$ flows to non-abelian or supersymmetric actions (for example ABJM with the 3d flow) could have larger impact. It is much harder, though, because flows of non-abelian DBI actions face ordering ambiguities. A systematic classification of decoupling limits for intersecting branes is easier but less decisive. The supergravity and symmetry test balances feasibility and impact best.